Barinder Singh Banwait

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Focus

I want to build the verification technology behind international AI agreements: the hardware, cryptography and formal verification that lets each side check the other is running inference and not a new training run, and learn nothing else. A collective slowdown requires this to work, and almost nobody is building it. Further out, I want provably safe AI in Tegmark's sense: systems that have passed a formal verification protocol, not a legal one. My background is formal proof in Lean 4, safety-critical embedded code under IEC 62304, and elliptic curves, the mathematics under elliptic-curve cryptography. This problem needs all three.

Constraints: London-based, UK citizen. Will relocate, including to the US: I was previously on a J-1 for three years at Boston University and MIT, and would need sponsorship again for anything long-term.

Target roles

Research scientist or research engineer in formal verification, hardware-enabled verification mechanisms, or compute governance. Teams on my list: ARIA Safeguarded AI, UK AISI, Oath, Attestable, CAISH's hardware assurance programme. Also open to technical governance research (GovAI, RAND) where verification is the research question.

Experience

- **Formal verification, Lean 4.** Complete machine-checked proof of the Ramanujan–Nagell theorem, solo (arXiv:2604.09808). Formal statement of the weak Birch–Swinnerton-Dyer conjecture, a Millennium Prize problem, merged into Google DeepMind's Formal Conjectures benchmark (PR #5371); strong form in progress, 7 PRs, 5 merged, defining objects Mathlib lacked. Lean formalisation of the main criterion of my latest paper (arXiv:2609.08431).
- **Safety-critical engineering.** Research Engineer, CMR Surgical, 2018–19. Designed the inverse-kinematics algorithm running on the Versius surgical robot (45,000+ patients since), in embedded C under MISRA C and IEC 62304. Technical lead on joint-fault prediction: managed 4 ML engineers, deployed on about 100 robots. I know what it takes to evidence a safety claim before hardware goes near a person.
- **ML and evaluations.** Built steelman_eval on Inspect: 25 contested issues, four-dimension rubric, cross-family judge, three frontier models. Proposed and led a 5-person project training neural networks on elliptic-curve data (ATMP 2025). Co-author on a decision-trees paper where the models suggested formulae we then proved (arXiv:2607.24251).
- **Research.** 17 papers in number theory, ML for mathematics and formal verification (full list). First generalisation of Mazur's 1978 isogeny theorem beyond the rationals: turned a non-effective proof into an algorithm, implemented in Sage, PARI/GP and Magma, solo (IMRN 2023); extended to all number fields (Algebra & Number Theory 2025). A year in LFANT, INRIA Bordeaux's algorithmic number theory and cryptography team (2014). BA/MMath Cambridge, PhD Warwick, positions at MIT, Boston University, Heidelberg.
- **Software.** LMFDB contributor 2020–25 (Python, Postgres). SageMath: first implementation of geometric-simplicity tests for genus-2 Jacobians (sage 9.5). Quant finance: mixed-integer optimisation of derivative portfolios (Gurobi).
- **Communication and leadership.** 60+ research talks, about 30 at international conferences. Led research groups at AIM, CIMPA Kampala, and now a 5-person KIAS group using frontier models on the generalised Fermat equation. Organised workshops from 20 to 200 people.

Engagement with AI safety

- BlueDot Impact AGI Strategy course, October 2026.
- On collective restraint (September 2026): where I disagree with point 4 of Yudkowsky's AGI Ruin, with the Asilomar conference as precedent.
- Read: AI 2027; Tegmark and Omohundro, Provably Safe Systems; Redwood on AI control; Alignment Faking; Gradual Disempowerment; Amodei, The adolescence of technology.
- Iliad Fellowship, London, November 2026 – February 2027: mentored technical AI safety research.